

Lab 1 - Food Link Description

Lab 1 Outline
FoodLink and the World Without
Green Team
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1. Introduction

Since the onset of the COVID-19 pandemic, the demand for food assistance in the United States has continued to rise, resulting in a growing national issue of food insecurity affecting millions of Americans. A large contributor to this nation-wide crisis is the inefficiency of our country’s food distribution system. According to ReFED’s 2023 data, The United States generated roughly \$388 billion worth of surplus food. Of this total, only about 2.3% of it was donated, while only 9% went to recycling, leaving the 88% remaining going to waste, leaving the already struggling individuals and families without reliable access to food despite the abundance of surplus resources.

Food Insecurity occurs when individuals lack consistent access to enough food necessary for a healthy daily life. While foodbanks serve as a critical safety net for millions of Americans, significant inconsistencies in their operations and distribution systems have created gaps in

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addressing this growing crisis. These inconsistencies have become particularly problematic as demand has surged due to rising living expenses, economic inflation, and recent funding cuts to SNAP and government programs.

According to data from Feeding America and the Virginia Peninsula Foodbank, approximately 11% of Virginia residents experience food insecurity, including 13.6% of children across the state. While some individuals may rely on well-funded facilities, others rely on small, under-resourced pantries or food banks with limited hours and irregular supply chains. These operational disparities mean that families facing identical levels of need may have vastly different access to food assistance depending on the food bank's operations, which in most cases, struggle with unpredictable donation flows,, and the lack of a cooperative networking technology.

One solution to this issue is an improved inter-foodbank coordination system. However, while resource sharing is a common procedure between foodbanks, it has proven to be very inconsistent and difficult due to the lack of coordination between foodbanks. Foodbanks operate with different inventory management systems, varying storage capacities, and unpredictable demand fluctuations. Additionally, foodbanks typically lack any information about neighboring facilities that may suffer from surpluses or shortages, causing a reactive resource sharing strategy rather than a consistent, proactive one. A viable solution must address all these issues in order to prevent these persistent issues.

A potential solution that aims to solve these problems is Food Link, a web-based application designed to create a unified network between foodbanks made for ease of transfers and reduced waste. It will accomplish these goals by creating an inventory system visible to other foodbanks, allowing facilities to identify available surpluses or urgent needs in real-time.

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Through a streamlined request and approval system, foodbanks can quickly coordinate transfers before predicted surpluses or shortages shown by the platform. By connecting isolated food banks together through this system, Food Link enables resources to be delivered to where they are needed, reducing waste and overstocking at locations with a lack of accommodation for storage, and providing more support to underserved communities.

2. Food Link Product Description

FoodLink will connect food bank inventory systems that are traditionally managed locally, and using a real-time database solution, provide a convenient means of coordinating the transit of resources between food banks. While FoodLink does not directly handle shipping logistics, the easy-to-use interface is designed to provide users with the ability to directly approve requests and coordinate transfers through a state-of-the-art, role-based authentication system that ensures privacy where users want it, and flexibility where users need it.

The software will lead to an overall increase in the benefit and utility of donations by ensuring they find their way to a home in need of support rather than spoiling while families in need go without. If this program is rolled out on a large scale, a decrease in the amount of waste generated would begin being seen immediately. Users in anticipation of high volunteer traffic could offer support between each other through the FoodLink interface and begin a coalition building process that leads to increased output of resources to those who need them.

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2.1. Key Product Features and Capabilities

FoodLink is designed with the users in mind. From an easy integration assistance program to jumpstart the process for users wondering if FoodLink is right for them to our 24/7/365 customer support and acquisition guarantee, FoodLink is built to empower the users with tools that allow them to communicate and network without the need for traditional meetings. By providing incentives for use, FoodLink is producing innovative strides in real-time food bank inventory networking. With options to track expiration and factor in transit time, FoodLink is able to ensure that the food is good all the way to its destination, but not in a way that implicates FoodLink if expired goods are transited within the network.

The biggest thing FoodLink does is aggregate resources between users to assess where more resources are needed. The current systems in place are what generate waste because food that could otherwise be used to support people is not known to people who are not physically at the food bank. By creating a network that incentivizes communication and coordination between users, FoodLink is expanding the good users are doing with a clean user-interface that spreadsheet software simply isn't able to provide. FoodLink provides users with a tool for expanding the good they do and maximizing the effective use of resources that have been donated..

2.2. Major Components (Hardware/Software)

2.2.1 Hardware

Web-Server

Database-Server

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User Device (PC, Laptop, Phone)

2.2.2 Software

Frontend: React.js

VSCode

Node.js

React Dev Tools

Tailwind

Backend: FastAPI

VSCode

Pytest

Git

Database: PostgreSQL

pgAdmin

3. Identification of Case Study

The primary case study for Food Link are food banks so that they can collaborate their inventory and other important information. This group was selected as Food Link is designed to be used by collaborating food banks in order to unite their distribution into a single powerful network.

Future users of Food Link could include different types of organizations such as schools or governments that could connect their information to provide aid

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4. X Product Prototype Description

Help decision-makers see inventory status across locations, split surplus/shortage conditions, and visualize sites on a map. To fit a one-semester build, we preserve the main user flows (log/update inventory, view status, map, export/report, respectfully) while reducing automation and ML scope, and simulating external feeds where needed. The result demonstrates feasibility of the data model, UI, and basic analytics while deferring production-grade scale, prediction, and routing logic to the RWP.

4.1. Prototype Architecture (Hardware/Software)

Implemented (prototype):

- *Inventory logging & manual updates*
- *Surplus/shortage rule-based alerts*
- *Location map via Google Maps API*
- *Basic reports (counts, summaries, CSV export)*

Simulated / Modeled:

- *“Real-Time” updates modeled with seeded data and scheduled refresh (5-15 min cron or manual refresh)*
- *Predictive alerts stubbed with placeholder scores to exercise UI states*

Reduced / Deferred to RWP:

- *Automated streaming ingestion and device/webhook integrations*
- *Route optimization and turn-by-turn suggestions*
- *Forecasting/trend analytics with model retraining and evaluation*

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- *Enterprise auth/SSO, hardening, and SLOs*

4.2. Prototype Features and Capabilities

In the prototype, inventory is updated through basic logging and periodic/manual refreshes, whereas the RWP automates real-time updates from external feeds and devices. Prototype alerts rely on simple, rule-based flags to identify surpluses and shortages. The RWP upgrades this to predictive alerts that forecast needs using demand and lead-time signals. Mapping in the prototype focuses on Google Maps visualization of sites, while the RWP adds routing suggestions and optimized multi-stop paths. Analytics are limited to basic status reporting and CSV exports in the prototype, but the RWP provides trend forecasts and richer time-series dashboards. Supporting pieces follow the same pattern, the prototype uses seed data/mock webhooks, local auth, and a single deployment, while the RWP introduces streaming/API ingestion with monitoring, SSO/MFA with fine-grained roles, and CI/CD with autoscaling, SLOs, and backups/DR.

4.3. Prototype Development Challenges

- *Scope & Architecture: Keeping a thin vertical slice while leaving room to expand later*
- *Data Quality: Seed/CSV data may be dirty or unrealistic. Need schema migrations, validation, and edge-case seeds.*
- *Alerts Logic: Simple rules are easy, Making thresholds configurable and explainable is harder*
- *Maps Integration: API keys/quotas and geocoding accuracy. Cache results and handle quota failures gracefully.*
- *AUTH & CORS: Minimal auth still requires session/CORS setup and coarse roles.*

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- *“Real-Time” Feel: Scheduled refresh can cause UI jitter. Use background refresh and small pages APIs*
- *DevOps Consistency: Different machines/environments. Solve with Docker, .env, and one-click seed/migrate*
- *Testing & Observability: Need basic unit/API tests and request logging to debug quickly.*

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5. Glossary

Database - an organized collection of information, or data, that can be easily accessed, managed, and updated electronically

Python - high-level, interpreted programming language known for its emphasis on code readability through significant indentation

PostgreSQL - a powerful, open-source, object-relational database management system

VSCode - open-source, and cross-platform source code editor developed by Microsoft

Git - open-source, distributed version control system used by developers to track changes in their code and manage projects over time

React.js - is a free and open-source front-end JavaScript library that helps build interfaces

User - References to the food bank

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